Data Visualization Project – Milestone 2

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I. Project Goal

Deforestation is a major environmental challenge, shaped by land use, economic development, and energy infrastructure. This project builds an interactive web visualization to explore the geographic relationship between power plant distribution and deforestation patterns, contextualized by population density.

The central question is: *Do regions with certain energy profiles coincide with higher deforestation pressure, and how does population density modulate this relationship?* To answer it, we overlay three global datasets on a shared map: the WRI Global Power Plant Registry [1], the GFW deforestation driver for 2001–2019 [2], and the EC JRC/CIESIN gridded population estimates for 2015 [3]. Together, they allow users to identify areas where energy infrastructure and environmental degradation intersect.

The target audience includes policymakers, environmental researchers, and data-literate citizens. The goal is to provide a clear, map-based understanding of which regions show the strongest co-occurrence of specific energy profiles and deforestation risk, to support more informed decisions around sustainable energy planning and forest conservation.

II. Visualization Sketches

Figure 1 presents the overall layout of our single-page interactive website. The interface is organized into three horizontal zones: a filter bar at the top, a central map area with a detail panel on the right, and a row of analytical charts at the bottom.

A. Filter Bar

A horizontal filter bar at the top of the page drives all views simultaneously. It exposes multiples dropdown menus (*Fuel Type*, *Country*, *Deforestation Driver*, ...) and two layer toggles (*Population Density* and *Deforestation Layer*). Any change propagates instantly to the map, the detail panel, and the charts below via D3 brush-and-link interactions.

B. Interactive World Map

The central element of the page is a Leaflet world map rendered using a natural-earth projection. Each power plant from the WRI Global Power Plant Registry is shown as a colored circle whose hue encodes the primary fuel type (e.g. coal in dark gray, hydro in blue, wind in green, solar in orange, nuclear in purple). Circle size can optionally encode installed capacity in MW. The deforestation layer (GFW raster, 2001–2019) is overlaid as a semi-transparent choropleth, color-coded by driver: commodity-driven deforestation, shifting agriculture, forestry, wildfire, and urbanization. The population density layer (EC JRC/CIESIN, 2015) can be toggled on as an additional heatmap. Users can zoom and pan freely; hovering over a plant reveals a tooltip with its name, fuel type, installed capacity, and the nearest deforestation driver. Clicking on a plant populates the detail panel on the right.

Fig. 1: Website layout sketch: filter bar (top), interactive world map with detail panel (center), and analytical charts (bottom).

C. Right Detail Panel

A fixed-width panel to the right of the map provides three widgets. The upper widget is a color legend mapping each fuel-type color to its label. The middle widget is a dynamic detail card: empty by default, it fills with the selected plant's attributes (name, country, fuel type, capacity, and local deforestation context) when the user clicks a map marker. A Planetary Boundaries reference block at the bottom of the panel situates the currently visible region within the broader framework of radiative forcing and HANPP limits.

D. Analytical Charts

Multiple charts are displayed side by side below the map, each updating in response to the active filters and the current map view. As of right now we display the total energy capacity of the map view, the powerplant count and the average capacity. All of these metrics are displayed by fuel type. In the final project more charts will be available. Additionally, the

user will be able to choose which charts he wants displayed either through a drop-down menu or other selection methods.

III. Tools and Lectures

A. Tools

The visualization is built as a single-page web application using the following tools and libraries.

a) D3.js: The core rendering library for all views: geographic projection, color-encoded power plant markers, and the three analytical charts. Further, it will use for the deforestation choropleth overlay, the population density heatmap. D3 also handles all brush-and-link interactions between the map and the charts, and is used to parse CSV files.

b) Leaflet / GeoJSON: Used for tile-based base map rendering and zoom/pan support. Deforestation and population raster data, preprocessed from TIF to GeoJSON using Python, are served as overlay layers on top of the Leaflet base map.

c) Python (Rasterio, GeoPandas, NumPy, Pandas): All preprocessing is done in Python: converting the GFW deforestation TIF and the EC JRC population TIF to CSV and GeoJSON, downsampling the population raster to $10\,\mathrm{km}^2$ resolution to match the deforestation grid, and adding reverse-geocoded country codes to each pixel.

d) HTML / CSS / JavaScript: The page layout, filter bar, toggle controls, and detail panel are implemented in vanilla HTML and CSS. JavaScript handles the event wiring between filter controls, the map, and the charts.

B. Relevant Lectures

- **D3.js & Interactions** (9–22 Mar): core rendering library for all charts and views, and the brush-and-link mechanism linking the filter bar, map, and charts.
- **Perception, Colors & Marks/Channels** (23–29 Mar): theoretical grounding for encoding fuel type as hue and installed capacity as circle size.
- **Maps** (20–26 Apr, upcoming): geographic projections and choropleth overlays, directly applicable to the deforestation and population layers.
- **Designing Viz** (13–19 Apr, upcoming): dashboard layout principles applied to our single-page design.

All course lectures contribute to the project, as the concepts introduced throughout the semester will collectively inform the design and implementation decisions made in this visualization.

IV. Project Breakdown

A. Core Visualization (Minimal Viable Product)

The following components constitute the core of the visualization and must be delivered at the end of the project:

1) **Interactive World Map**: display power plant locations as point markers, color-coded by primary fuel type, with zoom and pan support.
2) **Deforestation Overlay**: display deforestation data as a choropleth or raster layer that can be toggled on/off.
3) **Tooltip / Detail Panel**: on hover or click, show details about a selected power plant (name, country, fuel type, capacity).
4) **Filter Controls**: allow filtering power plants by fuel type and/or country.

B. Extra Ideas (Enhancements)

The following ideas could enrich the visualization but can be dropped without compromising the core message:

1) **Population Density Layer**: overlay population density data to explore the co-occurrence of energy infrastructure, deforestation, and human settlement.
2) **Linked Bar Chart**: a bar chart summarizing the fuel-type distribution for the currently visible map region, updated dynamically on pan/zoom.
3) **Correlation View**: a scatter plot or heatmap showing the statistical relationship between power plant capacity/type and deforestation intensity at the country or grid-cell level.
4) **Planetary Boundaries Panel**: a contextual sidebar situating the selected region's data within the planetary boundaries framework.

V. Functional Prototype

Our skeleton website is accessible at: https://com-480-data-visualization.github.io/Deforest-Gump/.
The current prototype includes:

- A HTML/CSS/JS skeleton implementing the page layout and filters described in Figure 1.
- An interactive map with a functional "Power Plants" layer. The two remaining layers (population density and deforestation) will be integrated in the next milestones.
- Three charts that update dynamically based on the current map view.

Note that the Power Plants layer currently does not wrap around when panning horizontally beyond the map boundaries, causing markers to disappear. This will be addressed in a future milestone.

References

[1] World Resources Institute, "Global power plant database," 2023. [Online]. Available: https://datasets.wri.org/dataset/globalpowerplantdatabase

[2] Global Forest Watch, "Global deforestation drivers 2001–2019," 2020. [Online]. Available: https://www.globalforestwatch.org

[3] European Commission JRC and CIESIN, "Global human settlement population grid 2015," 2015. [Online]. Available: https://ghsl.jrc.ec.europa.eu